

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	25 June 2026
Team ID	SWTID-2026-8141
Project Name	Heart Disease Analysis and Visualization Dashboard
Maximum Marks	4 Marks

The Deliverable shall include the architectural diagram below and the information as per Table-1 & Table-2.

Guidelines:

- Application Logic: The system uses HTML, CSS, JavaScript, and Python (Flask) for the web layer.
- Data Layer: Heart disease datasets are cleaned and structured using MySQL and SQL queries.
- Visualization Layer: Tableau Desktop is used to build interactive dashboards and stories, published via Tableau Public.
- Integration Layer: Tableau Public visualizations are embedded into the Flask web application for end-user access.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive web application	HTML5, CSS3, JavaScript
2	Application Logic-1	Flask application to host dashboard	Python, Flask
3	Application Logic-2	Dashboard visualization	Tableau Public
4	Application Logic-3	Data preprocessing and integration	Python, Pandas
5	Database	Structured heart disease datasets	MySQL / CSV Files
6	Data Source	Patient, hospital, and risk-factor datasets	CSV Dataset
7	File Storage	Project files and datasets	Local File System
8	External API-1	Tableau Dashboard Embedding	Tableau Public
9	External API-2	Dashboard/Story sharing	Tableau Public Embed API

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open Source Frameworks	Frameworks used in project	Flask, Bootstrap, Tableau Public
2	Security Implementations	Secure embedding through Tableau Public and GitHub repository, anonymized patient data	HTTPS
3	Scalable Architecture	Modular folder structure separating datasets, SQL, Tableau, screenshots, and documentation	Three-layer architecture
4	Availability	Dashboard accessible online 24x7 through Tableau Public and project available on GitHub	Tableau Public, GitHub

S.No	Characteristics	Description	Technology
5	Performance	Lightweight datasets with interactive Tableau visualizations	Tableau Public